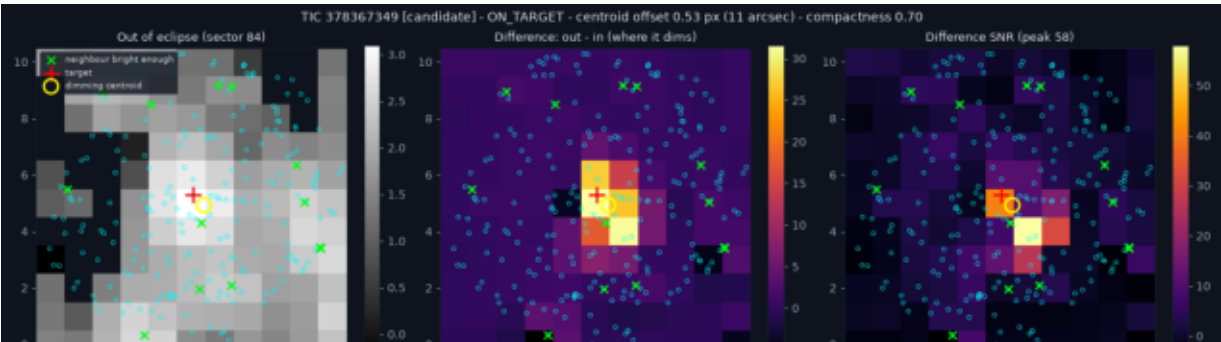
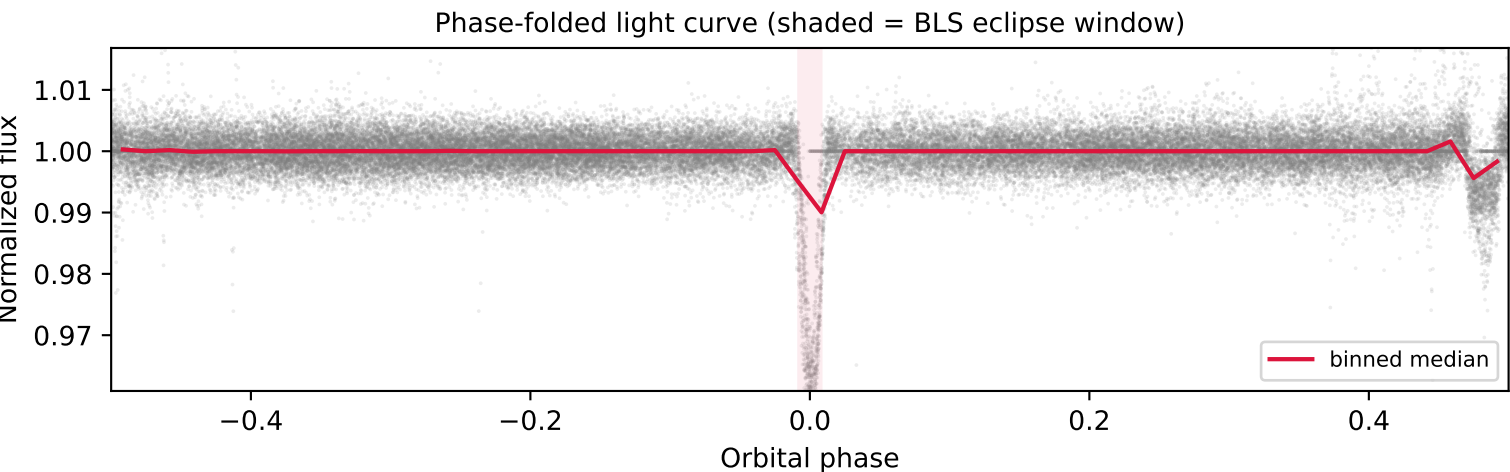


TIC 378367349

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	15.103695 d (ALIASED, see below)		
-> corrected period: 30.207391 d		VSX	no match
Epoch (t0)	1778.0515 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	2.040 % (20,402 ppm)	Gaia variable classification	no match
SNR	256.1	TESS EB catalog (Prsa+2022)	no match
		ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)			
Eclipses observed	12 (gate >= 3)	Verdict: NOVEL	
Robust depth ratio	1.37 (gate >= 0.50)		
Robust depth sigma	249.1 (gate >= 7.0)		
Odd/even z-score	12.58 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase 0.48, 6.9 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 84	light-curve dip in this sector: 129.0 sigma	t0 refined by 0.0025 phase	
Difference-image peak SNR	58.11 (gate >= 5.0)	Compactness 0.70 (gate >= 0.50)	
Dimming centroid offset	0.53 px (11 arcsec)	(gate <= 1.0 px, i.e. on the target)	
Nearest bright-enough neighbour in cutout: TIC 378367373			
0.62 px away, dTmag +1.28 - farther from the dimming than the target			



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 12.58 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel check confirmation and eclipse timing are unaffected - report the corrected period, 30.207391 d, in any write-up. Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.085 px (dimming centroid 0.534 px from target vs 0.619 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.